AAT 2

Statistical Machine Learning

Assessment Pattern

Component	Marks	Assessment Criteria
Project Execution	5	Implementation, data handling, analysis, graphs/results,and demonstration
Project Report	5	Problem, objectives, methodology, results, interpretation ,and conclusion
Viva Voce	5	Conceptual understanding, implementation, results

Project Requirement

Students should complete a simple practical project related to Statistical Machine Learning. The project should demonstrate the application of a statistical or machine-learning concept to a small dataset or simulated dataset. Students should be able to execute the project, explain their methodology and results, and answer questions during the viva.

Example Topics

1. A/B Testing Analysis
Compare two versions of a webpage, advertisement, product, or learning activity. Calculate response rates, formulate hypotheses, and perform statistical analysis.

2. Permutation Test Simulation
Generate two groups of data, calculate the observed difference, perform repeated permutations, plot the permutation distribution, and estimate the p-value.

3. Multi-Armed Bandit Simulation
Create 3–4 arms and implement ε-greedy to study exploration, exploitation, and cumulative reward.

4. UCB-Based Advertisement Selection
Simulate multiple advertisements and use UCB to select arms while tracking rewards and selections.

5. Thompson Sampling Simulation
Use Beta distributions for binary rewards and simulate arm selection and distribution updates.

6. Comparison of Bandit Algorithms
Implement ε-greedy, UCB, and Thompson Sampling on the same problem and compare their results.

Project Report Format

1. Title
2. Problem Statement
3. Objectives
4. Dataset / Data Generation
5. Methodology / Algorithm
6. Implementation
7. Results – tables, graphs, and important calculated values
8. Conclusion

Marking Scheme

Project Execution – 5 Marks	**Marks**
Correct implementation	2
Proper data handling / calculation	1
Results / graphs	1
Successful demonstration	1
Project Report – 5 Marks	**Marks**
Problem statement & objectives	1
Methodology	1
Implementation & results	2
Interpretation & conclusion	1
Viva Voce – 5 Marks	**Marks**
Conceptual understanding	2
Explanation of implementation	1
Questions / cross-questioning	2
TOTAL	**15**

Submission Instructions

• The project should be simple, practical, and related to Statistical Machine Learning.
• Students may use Python, Excel, or another suitable tool for implementation.
• Students should be prepared to demonstrate the working project during evaluation.
• Students should understand their own implementation and results, as the same will be assessed during the viva.